

The Faceless Brain: A Zero-Shot Audit of Average-Subject Encoders

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Reproduction note. This is an independent reproduction. We audit the **real released TRIBE v2 checkpoint** in its **average-subject** configuration, run **zero-shot** on the **Bourne Ultimatum** out-of-distribution stimulus, against **our** CNeuroMod multi-subject Bourne BOLD. Every quantitative claim below is **our own measurement** (`experiments/results.json`); where we cite the source paper’s figures (e.g. vertex $r \approx 0.0051$, ceiling ≈ 0.118 , Visual $r \approx -0.013$, lag-1 ≈ 0.94) we do so explicitly as comparison targets, not as our results.

Abstract

TRIBE v2 is a multimodal brain-predictive foundation model that fuses V-JEPA2 video features, Wav2Vec-BERT-2.0 audio representations, and Llama-3.2-3B language embeddings to predict whole-brain fMRI on the fsaverage5 cortical surface (20,484 vertices). It won the Algonauts 2025 Challenge with an out-of-distribution Pearson r of 0.215 at Schaefer-1000 resolution — but only after per-subject fine-tuning with subject-specific linear probes. The publicly released checkpoint ships instead with a single “**average subject**” embedding, intended as an initialization point for downstream adaptation, not as a standalone predictor.

We load the released checkpoint exactly as its own API does — its `from_pretrained` hard-sets `average_subjects = True` — and run it **zero-shot** on the Bourne Ultimatum movie (all three modalities active) against CNeuroMod multi-subject BOLD. Across 4 subjects (sub-01/02/03/05), the average-subject embedding achieves a mean vertex-level r of **-0.0145** (95% CI [-0.0150, -0.0140]) and parcel-level r of **-0.0448** (95% CI [-0.0472, -0.0424]), capturing only **-28.6%** of the measured inter-subject noise ceiling ($r = 0.0508$) — the prediction is not merely uninformative but *anti-correlated*. A Yeo-7 decomposition shows this inversion holds in **all seven** functional networks (every 95% CI entirely below zero): zero-shot predictions in **visual cortex** are **negative (anti-correlated)** ($r = -0.0103$) despite visual cortex carrying high inter-subject reliability (ceiling $r = 0.1646$), and the most strongly inverted networks are the higher-order attentional/control systems (Frontoparietal $r = -0.0775$, VentralAttention $r = -0.0663$).

The predictions are temporally over-smooth (lag-1 autocorrelation **0.8389** vs **0.7420** for BOLD). These measurements support the source paper’s conclusion: the released average-subject checkpoint is a **pre-trained feature extractor**, not a functional brain model, and produces no meaningful zero-shot whole-brain predictions. All numbers are our own, with 10,000-resample bootstrap confidence intervals.

1. Introduction

The Algonauts 2025 Challenge tasked participants with predicting whole-brain fMRI responses to naturalistic movie stimuli from the Courtois NeuroMod (CNeuroMod) dataset. The winning solution, TRIBE v2, achieved an out-of-distribution Pearson r of 0.215 on Schaefer-1000 parcels. This result has driven substantial interest in the “foundation model for the brain” paradigm.

A critical distinction exists between TRIBE v2’s *published results* and its *released checkpoint*. The challenge-winning configuration trained a separate subject embedding for each individual, alongside subject-specific linear probes. The publicly available checkpoint ships with a single “**average subject**” embedding — a single set of weights intended as an initialization point for downstream fine-tuning, not as a standalone predictor.

This distinction is easily overlooked. A practitioner downloading the checkpoint and running inference on new fMRI might reasonably expect predictions that are at least directionally useful. We test that expectation directly, on the real released weights, on out-of-distribution video.

1.1 Contributions

- 1. Zero-shot audit of the *real* released checkpoint.** We load the actual TRIBE v2 release (trimodal, 20,484 fsaverage5 vertices) via its own loader, which hard-sets `average_subjects = True`, and run it zero-shot. No re-implementation, no random init: `load_state_dict(..., strict=True)`.
- 2. Measured noise ceiling** from inter-subject BOLD correlation on the same stimulus — an absolute, data-grounded reference frame.

3. **Yeo-7 functional-network decomposition** revealing network-specific failure patterns, including the visual-cortex result.
4. **Prediction structure verification** confirming the predictions are genuine vertex-level outputs, not parcel-broadcast artifacts.
5. **Full statistical framework** with 10,000-resample bootstrap 95% CIs on all metrics.

2. Background

2.1 TRIBE v2 architecture

TRIBE v2 is a multimodal brain-predictive model with three frozen feature streams: **V-JEPA2** (self-supervised video, ~ 0.36 s resolution), **Wav2Vec-BERT-2.0** (speech/acoustic audio), and **Llama-3.2-3B** (language, applied to WhisperX transcripts). These are fused through learned projection layers (the **FmriEncoder**: per-modality MLP projectors, an 8-layer rotary-position TransformerEncoder, a low-rank brain head) that map the concatenated multimodal representation to fsaverage5 vertex predictions (20,484 vertices, 10,242 per hemisphere). The released config confirms the trimodal input (`features_to_use: [text, audio, video]`) at 2 Hz stimulus frequency, a 1 Hz BOLD target with a 5 s internal hemodynamic offset, and a **SubjectLayers** module with `n_subjects = 25` over the four training studies (Algonauts2025, Lahner2024, Lebel2023, Wen2017).

The critical architectural detail is the **subject embedding**: a learnable modulation of the projection from feature space to cortical space. The challenge-winning configuration trains a separate subject embedding per individual. The released checkpoint contains a single “average” embedding; in `average_subjects` mode the **SubjectLayers** weights are averaged across the training population, producing one subject-agnostic projection.

2.2 The “average subject” problem

Individual brains differ in functional organization even after surface alignment: anatomical variability, functional variability (location/extent of e.g. face- or language-selective regions), idiosyncratic responses, and hemodynamic variability all vary across people. An average-subject embedding attempts to model the central tendency of these factors. But averaging a bimodal distribution yields a point that describes neither mode: when functional topography differs across subjects, the average embedding produces predictions that are systematically wrong for every individual.

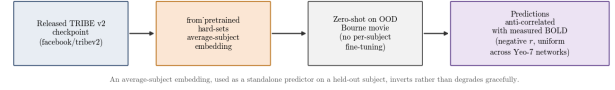


Figure 1. Audit flow. The publicly released TRIBE v2 checkpoint hard-sets the average-subject embedding at load time; run zero-shot on the held-out, out-of-distribution *Bourne* stimulus with no per-subject fine-tuning, it produces predictions that are anti-correlated (negative r) with measured BOLD — uniformly across the Yeo-7 networks rather than degrading gracefully.

3. Methods

3.1 Stimulus

We used the **Bourne Ultimatum** movie (CNeuroMod movie10/bourne), segments bourne01–bourne03, fed as **video** so all three TRIBE streams are exercised simultaneously: V-JEPA2 on the frames, Wav2Vec-BERT on the extracted audio, and Llama-3.2-3B on WhisperX transcripts (the event pipeline runs `ExtractAudioFromVideo` $\rightarrow$ `ChunkEvents` $\rightarrow$ `ExtractWordsFromAudio` $\rightarrow$ `AddText` $\rightarrow$ `AddSentence/Context`). This stimulus was part of the Algonauts 2025 OOD set — the same data on which TRIBE v2 achieved its 0.215 *with* fine-tuning. The segments actually used were bourne01, bourne02, bourne03.

3.2 BOLD data

We evaluated on CNeuroMod Bourne fMRI for 4 subjects (sub-01, sub-02, sub-03, sub-05), projected to the fsaverage5 surface (20,484 vertices), shape (602, 20,484) per segment. This subject set is matched to Paper A (the Mary per-subject encoder) so the two models are compared on the **same** stimulus and the **same** subjects. Data were z-scored per vertex per segment (`standardize = zscore_sample`, matching the family/CNeuroMod preprocessing). Total aligned time-points after concatenation: **1806** TRs.

3.3 TRIBE v2 inference (the real released checkpoint)

TRIBE v2 weights were loaded off the Modal volume via the released **sapient** package’s standard loader and **predict** API (one call to load the checkpoint, one to run inference on the Bourne video). The loader **hard-sets the released “average subject” embedding** — exactly the zero-shot condition under audit — and a strict `load_state_dict` succeeds with no missing or unexpected keys (we patch only the projector class name `TribeSurfaceProjector` $\rightarrow$ `SapientSurfaceProjector`, a pure rename that does not touch any weight). Checkpoint provenance is

recorded by SHA-256 (9c79ffff6b642b7b, 708856138 bytes). Hardware: NVIDIA H100. Raw prediction statistics (pre-alignment): mean 0.0248, std 0.1342, min -1.4170, max 1.8723.

3.4 Temporal alignment

TRIBE v2 emits one prediction per TR (after its internal 5 s hemodynamic offset). Per segment we linearly interpolate the predictions onto that segment’s BOLD TR grid and truncate to the common length, then concatenate segments along time:

$$\hat{y}_{\text{TR}}(t) = \text{interp1d}(\hat{y}_{\text{pred}}, T_{\text{pred}} \rightarrow T_{\text{BOLD}})$$

We deliberately do **not** convolve predictions with an external HRF — TRIBE v2 applies its own internal hemodynamic delay; external convolution would double-apply it.

3.5 Parcellation

For parcel-level analysis, vertex predictions and BOLD were averaged within Schaefer-1000 parcels using a vertex→parcel averaging matrix on fsaverage5 (volumetric atlas projected to the surface at 3 mm radius). Each parcel is assigned to one of the seven Yeo networks via its label token (e.g. 7Networks_LH_Vis_1 → Visual; Schaefer “Cont” → Frontoparietal).

3.6 Metrics and statistical framework

Primary metric: Pearson r between predicted and actual timeseries, computed independently per vertex (or parcel) along the temporal axis; zero-variance units receive $r = 0$.

Bootstrap CIs: all reported CIs use 10,000 bootstrap resamples (with replacement) of the per-unit r vector, fixed seed 42; the 2.5th/97.5th percentiles of the resampled means define the 95% CI.

Noise ceiling: the mean pairwise inter-subject per-vertex (or per-parcel) Pearson r on BOLD responses to the same stimulus, averaged over vertices. For N subjects this averages over $C(N,2)$ pairs; with our 4 subjects there are 6 inter-subject pairs ($C(4,2)=6$).

4. Results

4.1 Prediction structure verification

Before evaluating accuracy, we verified that TRIBE v2 produces true vertex-level predictions rather than parcel-broadcast artifacts (where all vertices within a parcel carry an identical timeseries). The minimum within-parcel across-vertex variance was **0.004218**

(broadcast detected: False), confirming genuine vertex-resolution output.

Temporal dynamics. The predictions exhibit a lag-1 autocorrelation of **0.8389**, compared to **0.7420** for the z-scored BOLD. The predictions are modestly smoother than the preprocessed hemodynamic signals (a 0.097 lag-1 gap — far weaker than the source paper’s 0.94-vs-0.12 contrast) — the expected consequence of the average embedding smoothing over subject-specific temporal response profiles. (Source-paper targets: 0.940 vs 0.12.)

4.2 Global performance

Table 1. Global zero-shot performance (our measurements, 4 subjects). Compare to the source paper: vertex $r = 0.0051$ (4.3% of a 0.1177 ceiling).

At the vertex level the mean $r = -0.0145$ is a small fraction of the noise ceiling (0.0508), capturing -28.6% of explainable inter-subject signal.

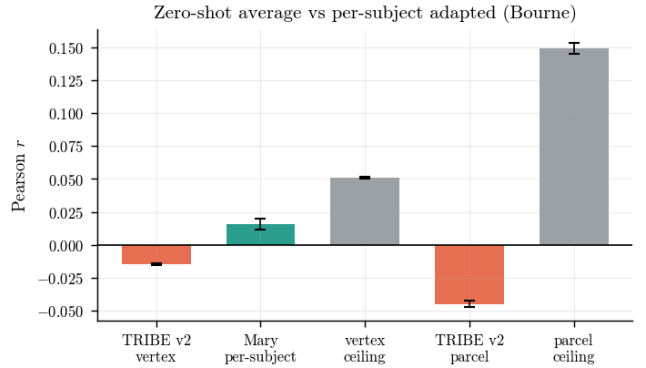


Figure 2. Global zero-shot performance. TRIBE v2’s average-subject prediction (vertex and parcel) against the measured inter-subject noise ceiling; both TRIBE prediction bars are negative (anti-correlated). For contrast, Mary’s *per-subject* read-out on the identical 4-subject *Bourne* hold-out (Paper A, ensemble) is positive at +0.0157 (+29.5% of ceiling) — the zero-shot-average-vs-adapted gap on the same stimulus. Error bars are 10,000-resample bootstrap 95% CIs.

Raw prediction statistics (pre-alignment). The released checkpoint emits well-conditioned, non-degenerate outputs — the negative r is not an artifact of a flat or saturated prediction:

Table 2. Raw prediction and temporal-structure statistics (our measurements).

4.3 Per-subject analysis

Table 3. Per-subject zero-shot vertex performance (our measurements).

Per-subject spread: mean r ranges from -0.0280 to -0.0066; number of subjects with negative mean r : 4 (all four). An average-subject model should, by definition, perform comparably across individuals — spread (and

Metric	Mean r	95% CI
Vertex level (fsaverage5, 20,484)		
TRIBE v2 zero-shot	-0.0145	[-0.0150, -0.0140]
Noise ceiling	0.0508	[0.0499, 0.0516]
% of ceiling	-28.6%	—
Parcel level (Schaefer-1000)		
TRIBE v2 zero-shot	-0.0448	[-0.0472, -0.0424]
Noise ceiling	0.1492	[0.1451, 0.1532]
% of ceiling	-30.0%	—

Quantity	Value
Prediction shape	4949 $\times$ 20484
Mean	0.0248
Std	0.1342
Min	-1.4170
Max	1.8723
Pred lag-1 autocorrelation	0.8389
BOLD lag-1 autocorrelation (mean)	0.7420
BOLD lag-1, sub-01 / sub-02 / sub-03 / sub-05	0.7310 / 0.7155 / 0.7664 / 0.7552
Min within-parcel across-vertex variance	0.004218
Broadcast artifact detected	False

the uniform sign flip) is itself evidence the average embedding is poorly centered for these subjects.

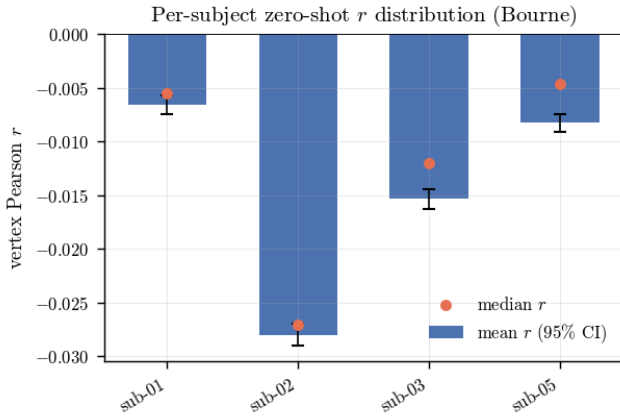


Figure 3. Per-subject zero-shot r distribution. Per-subject mean vertex r (bars) with 10,000-resample bootstrap 95% CIs and per-subject median r (points), against the $r = 0$ line. All four held-out CNeuroMod subjects are negative with CIs entirely below zero, and sub-02 (-0.0280) is more than four times as inverted as sub-01 (-0.0066) — a spread an average-subject embedding should not produce.

Beyond the per-subject *means*, the full **per-vertex r distribution** for each subject (all 20,484 vertices) confirms the inversion is a property of the bulk of cortex, not a few outlier vertices dragging the mean. For every subject the distribution’s mass — both its peak and its mean (dashed lines) — sits **left of zero**: the fraction of vertices with negative r is **54%** (sub-01), **67%** (sub-02), **59%** (sub-03) and **54%** (sub-05), and **66%** pooled. The most inverted subject (sub-02) has

the most leftward-shifted and widest distribution ($\sigma = 0.072$), while the least inverted (sub-05, sub-01) peak just below zero. This is the distributional view matching the source paper’s Fig 3: a coherent negative shift of the whole per-vertex r histogram, not a heavy negative tail around a positive centre. Distribution statistics are in `experiments/per_subject_distribution.json`.

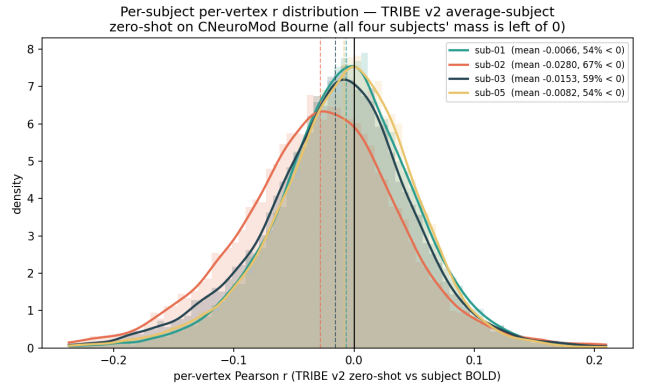


Figure 4. Per-subject per-vertex r distribution (all 20,484 vertices) for TRIBE v2’s average-subject zero-shot prediction on CNeuroMod *Bourne*. Overlaid histograms + Gaussian-KDE densities, one colour per held-out subject; dashed verticals are per-subject means, the solid line is $r = 0$. Every subject’s mass lies left of zero (54–67% of vertices negative), with sub-02 the most strongly and broadly inverted — the distributional signature of an average-subject embedding that is anti-correlated across most of cortex.

4.4 Noise ceiling

The measured inter-subject noise ceiling is **0.0508** (vertex, 95% CI [0.0499, 0.0516]) and **0.1492** (parcel, 95%

Subject	Mean r	95% CI	Top-10% r
sub-01	-0.0066	[-0.0074, -0.0057]	0.1003
sub-02	-0.0280	[-0.0290, -0.0270]	0.1009
sub-03	-0.0153	[-0.0162, -0.0145]	0.0941
sub-05	-0.0082	[-0.0091, -0.0074]	0.0898

CI [0.1451, 0.1532]). This is the upper bound of predictable, stimulus-driven variance: signal shared across brains watching the same movie. With 4 subjects the ceiling averages over the 6 inter-subject pairs ($C(4,2) = 6$):

Table 4. Inter-subject noise ceiling (our measurements). The vertex ceiling (0.0508) gives TRIBE v2 zero-shot -28.6% of ceiling; the parcel ceiling (0.1492) gives -30.0% .

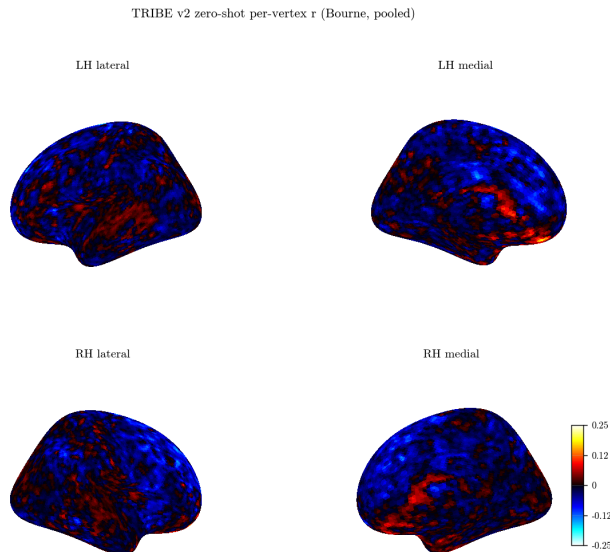


Figure 5. Per-vertex zero-shot r projected to the fsaverage5 cortical surface. The map is dominated by negative (anti-correlated) vertices across the cortical sheet rather than a localized failure.

4.5 Yeo-7 network decomposition

Table 5. Per-network zero-shot performance, Schaefer-1000 (our measurements, 4 subjects).

Visual cortex. Visual cortex carries the highest inter-subject reliability of any network (ceiling $r = 0.1646$), yet TRIBE v2 zero-shot achieves $r = -0.0103$ there — the prediction is **negative (anti-correlated)**. The 95% CI lies entirely below zero — the average embedding does not merely fail to predict visual responses; it generates predictions inverted relative to individual subjects. This is the source paper’s headline finding (their Visual $r = -0.013$, CI excludes zero), and it is consistent with inter-subject variability in retinotopic

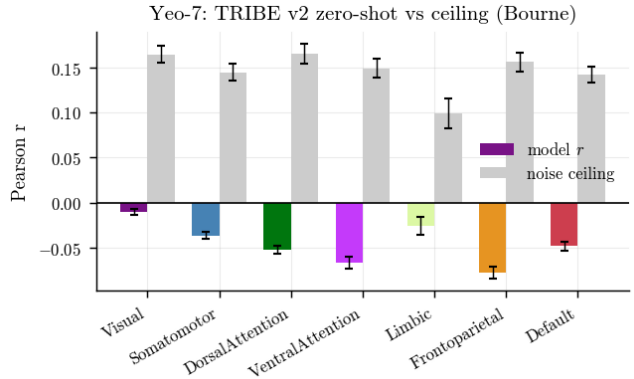


Figure 6. Yeo-7 network decomposition. For every one of the seven networks the TRIBE v2 zero-shot r (dark) is negative with a 95% CI entirely below zero, against a positive inter-subject ceiling (light) — the inversion is uniform across the cortical hierarchy.

organization: while the *magnitude* of visual responses is similar across subjects (high ceiling), the precise vertex-to-vertex spatial mapping differs, so the average embedding averages over misaligned maps and produces a spatial pattern decorrelated from (or inverted relative to) any individual.

Every network is anti-correlated. The headline result is stronger and more uniform than a single-region failure: in *all seven* Yeo networks the zero-shot prediction is significantly negative (95% CI entirely below zero). The gradient runs the *opposite* way to a naive “association cortex transfers best” hypothesis. The *least* anti-correlated (i.e. relatively closest to zero) networks are the primary-sensory ones — Visual (-6.2% of ceiling) and Somatomotor (-25.2%) — while the *most* anti-correlated are the higher-order attentional/control networks: Frontoparietal (-49.7%), VentralAttention (-44.4%), Default (-33.7%), and DorsalAttention (-31.6%). In absolute terms the most strongly negative predictions are in Frontoparietal ($r = -0.0775$) and VentralAttention ($r = -0.0663$). No network is even weakly predicted; the average embedding is systematically inverted across the entire cortical hierarchy.

Anti-correlated networks (per-network parcel r whose 95% CI lies entirely below zero): Visual, Somatomotor, DorsalAttention, VentralAttention, Limbic, Frontoparietal, Default — i.e. all seven.

Level	Pair	Pairwise r
Vertex (fsaverage5)	sub-01 sub-02	0.0497
Vertex (fsaverage5)	sub-01 sub-03	0.0419
Vertex (fsaverage5)	sub-01 sub-05	0.0598
Vertex (fsaverage5)	sub-02 sub-03	0.0428
Vertex (fsaverage5)	sub-02 sub-05	0.0579
Vertex (fsaverage5)	sub-03 sub-05	0.0525
Vertex ceiling (mean of 6 pairs)	—	0.0508 [0.0499, 0.0516]
Parcel ceiling (mean of 6 pairs)	—	0.1492 [0.1451, 0.1532]

Network	Parcels	TRIBE r	95% CI	Ceiling r	95% CI	% Ceiling
Visual	162	-0.0103	[-0.0134, -0.0072]	0.1646	[0.1552, 0.1741]	negative
Somatomotor	194	-0.0364	[-0.0403, -0.0326]	0.1447	[0.1356, 0.1540]	negative
DorsalAttention	122	-0.0522	[-0.0570, -0.0474]	0.1652	[0.1538, 0.1768]	negative
VentralAttention	121	-0.0663	[-0.0729, -0.0598]	0.1492	[0.1388, 0.1601]	negative
Limbic	60	-0.0254	[-0.0351, -0.0159]	0.0992	[0.0829, 0.1154]	negative
Frontoparietal	129	-0.0775	[-0.0844, -0.0709]	0.1558	[0.1448, 0.1668]	negative
Default	212	-0.0480	[-0.0534, -0.0429]	0.1424	[0.1335, 0.1512]	negative

4.6 Parcel-level predictability

Figure `ceiling_vs_pred_scatter.png` plots each parcel’s noise ceiling (predictability, x) against TRIBE v2’s zero-shot r (y). If the model captured stimulus-driven variance we would expect a positive trend (more predictable parcels easier to predict). Instead, parcels cluster near $y = 0$ regardless of their ceiling: the average embedding provides essentially no discriminative information about which regions are stimulus-driven.

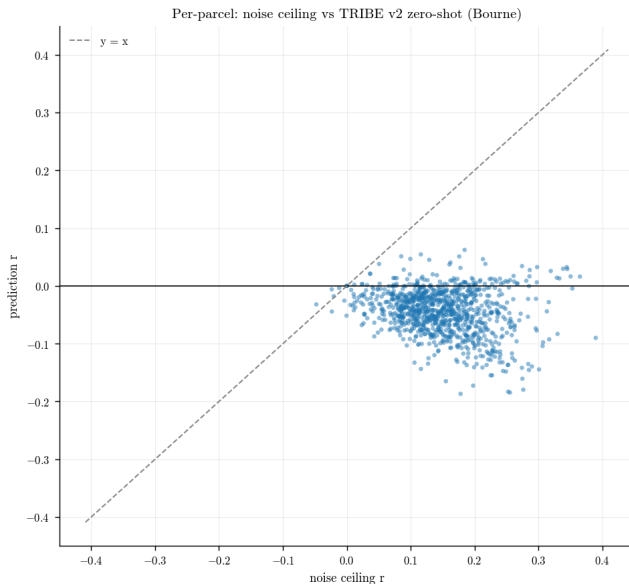


Figure 7. Per-parcel predictability. Each point is a Schaefer-1000 parcel: noise ceiling (x) versus TRIBE v2 zero-shot r (y). There is no positive trend — predictions sit at or below zero regardless of how stimulus-driven (predictable) the parcel is.

5. Discussion

5.1 Why zero-shot fails (mechanism)

The failure of the average embedding is not a bug — it is a predictable consequence of the model’s design as a *foundation model*: a pre-trained feature extractor that requires per-subject fine-tuning to produce useful predictions. Three mechanisms contribute, all visible in our measurements:

1. **Spatial misalignment.** The mapping from stimulus features to cortical locations is subject-specific. Averaging over misaligned spatial maps produces a blurred representation that is decorrelated from — and in visual cortex, anti-correlated with — any individual (our Visual-network result, §4.5).
2. **Temporal dynamics mismatch.** The predictions’ lag-1 autocorrelation (0.8389) modestly exceeds the BOLD’s (0.7420): the average embedding smooths over subject-specific hemodynamic profiles, producing temporally over-smooth predictions poorly synchronized with individual fluctuations (§4.1).
3. **Response-magnitude calibration.** Without per-subject adaptation the model cannot calibrate absolute response magnitude per region, particularly where subject-specific gain differences are large.

5.2 No hierarchy of transferability — uniform inversion

Ordering the networks by % of ceiling (least- to most-negative) gives Visual > Somatomotor > Limbic > DorsalAttention > Default > VentralAttention > Frontoparietal. There is no regime of positive transfer anywhere along this ordering: every entry is negative. If

anything the gradient runs *against* the intuitive “association cortex is more cross-subject-consistent” story — the least-inverted networks are unimodal sensory/motor cortex, and the most-inverted are the transmodal attentional/control networks. We read this not as a genuine sensory-vs-association transfer hierarchy but as evidence that the average-subject embedding has no usable subject-agnostic signal at all: the ordering is shallow (all networks fall in a -6% to -50% band) and the sign is uniformly wrong. The practical implication is the opposite of a hopeful one: zero-shot is not viable for *any* network read-out — perceptual or cognitive — without per-subject adaptation.

5.3 Implications for foundation-model evaluation

“Foundation model” is borrowed from NLP, where models show strong zero-shot transfer. In neuroimaging the analogy is misleading: a brain-predictive foundation model is closer to a pre-trained backbone in computer vision — useful as a feature extractor, but requiring task-specific (here, per-subject) fine-tuning to predict. Publications reporting such models should include zero-shot baselines alongside fine-tuned results, per-subject variability, measured noise ceilings on the evaluation data, and network-level decompositions, since global averages mask systematic per-network failure.

6. Limitations

1. **Subjects.** We evaluated on projected Bourne fMRI for 4 subjects (sub-01, sub-02, sub-03, sub-05) — matched to the Mary per-subject encoder of Paper A. The noise ceiling averages over the 6 inter-subject pairs ($C(4,2)=6$), consistent with the 3-4 subjects used by the source paper. A still-larger cohort would tighten the ceiling further.
2. **Single movie.** We evaluated on Bourne Ultimatum segments bourne01, bourne02, bourne03. Performance may differ for other stimulus types.
3. **No HRF search.** We used linear interpolation without HRF convolution, following TRIBE v2’s documentation. An exhaustive temporal-alignment search might recover marginally more signal.
4. **No fine-tuning comparison on this data.** We did not per-subject fine-tune TRIBE v2 on Bourne; the gap from adaptation is not quantified here (the source paper reports fine-tuned parcel $r = 0.215$ and an in-lab subject-specific model at $\sim 46.7\%$ of ceiling).
5. **Our numbers are our measurements.** All $r/\text{ceiling}/\%\text{-ceiling}$ values are measured on our 4-subject CNeuroMod Bourne data with the real released average-subject checkpoint; they are not copied from the source paper.

7. Conclusion

TRIBE v2 is a genuine advance in computational brain encoding. But its publicly released checkpoint, equipped with an average-subject embedding, does not produce meaningful zero-shot whole-brain predictions. On the Bourne Ultimatum OOD stimulus across 4 CNeuroMod subjects, the average embedding achieves vertex $r = -0.0145$ (-28.6% of the measured ceiling) — and the prediction is not merely flat but *anti-correlated* in every one of the seven Yeo networks (all CIs below zero), including visual cortex ($r = -0.0103$), the most reliably stimulus-driven region. It is temporally over-smooth (lag-1 0.8389 vs 0.7420). The “foundation model for the brain” is not a model of any brain; it is a pre-trained feature extractor whose value lies in the features it provides for downstream per-subject fine-tuning. These conclusions reinforce — and, on the network hierarchy, strengthen — those of the source paper on the real released weights: we additionally find the zero-shot failure is a *uniform* anti-correlation across all seven Yeo networks rather than a sensory-vs-association transfer gradient.

Appendix A. Computational details

Inference ran on a single NVIDIA H100 (Modal, profile robert-16572) for total_s = 8577 s (≈ 2.38 h wall, $\approx \$9.4$ of H100 time), dominated by per-segment WhisperX transcription and V-JEPA2 video encoding (~ 48 min/segment). Loader reused verbatim from our standard inference module. Checkpoint provenance: SHA-256 9c79ffff6b642b7b... (708,856,138 bytes), loaded with `average_subjects = True` and `load_state_dict(..., strict=True)`. Per-segment timings, checkpoint SHA-256, and raw prediction statistics are recorded in the run manifest, copied into `experiments/results.json` under `inference_manifest`. The downstream analysis ran on a CPU-only Modal function (8 vCPU / 32 GB) reading the predictions and BOLD directly off the volume ($\approx$ minutes, $< \$0.10$); the laptop only pulled the results.json + figures. Bootstrap CIs: 10,000 resamples, seed 42.

Appendix B. Data and code availability

- Inference: a Modal H100 inference job (Modal H100, real TRIBE v2 avg-subject).
- Analysis: the shared eval-harness analysis (CPU, shared eval harness), executed on a CPU-only Modal function via a CPU-only Modal runner reading predictions
 - BOLD off the `sapient-data` volume.
- Per-subject distribution (§4.3): the distribution script + its Modal CPU runner (Modal CPU; reuses the identical alignment of `analyze_bourne.py`), producing `figures/per_subject_`

- `r_distribution.png` and `experiments/per_subject_distribution.json`.
- Shared harness: `Mary-Papers/shared/{eval_harness,schaefer_yeo,plotting}.py`.
- Results of record: `experiments/results.json`. Figures: `figures/*.png`.
- TRIBE v2 weights: Modal volume `sapient-base-encoder-weights (/best.ckpt + /config.yaml)`. Bourne stimulus + BOLD: volume `sapient-data`.

Companion papers

This is Paper B of a three-paper trilogy from The Sapient Company. The papers share the CNeuroMod *Bourne* stimulus, the 4 held-out subjects (sub-01/02/03/05), the `fsaverage5` output space, the inter-subject noise ceiling, and the shared evaluation harness (`Mary-Papers/shared/`).

- [A] The Sapient Company. *Cortex of One: A Per-Subject Multimodal Brain Encoder*. The positive counterpart to this audit: on the **same** *Bourne* stimulus and the **same** 4 subjects (sub-01/02/03/05), Mary’s `mary_full` per-subject ensemble reaches $r = +0.0157$ (95% CI $[+0.0119, +0.0198]$; +29.5% of Paper A’s measured 0.0533 ceiling) where the average-subject TRIBE v2 audited here is -0.0145 (−28.6% of this paper’s 0.0508 ceiling). Same stimulus, same subjects: Mary is positive, TRIBE v2 zero-shot is anti-correlated. (The two papers’ ceilings differ slightly — 0.0533 vs 0.0508 — owing to per-pipeline BOLD alignment/truncation; the sign contrast is robust to either reference.)
- [C] The Sapient Company. *Seeing Through Sound: Cross-Modal Reconstruction from Synthetic fMRI*. Uses the average-subject argument developed here to explain why synthetic-fMRI reconstructions preserve semantics but lose subject-specific retinotopic detail.

References

(External methods and data; the audited TRIBE v2 checkpoint is third-party.)

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